

Lab Week 11: Social Engineering

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Google Dorking

Explaining how Search Engines work and leveraging them into finding hidden content!

45 min 150,618

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Video

Google Dorking | DarkStar • Jan 17, 2021

Source: YouTube

 TryHackMe Google Dorking Official Walkthrough

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Task 1  Ye Ol' Search Engine

Google is arguably the most famous example of "Search Engines", I mean who remembers Ask Jeeves? *shudders*

Now it might be rather patronising explaining how these "Search Engines" work, but there's a lot more going on behind the scenes than what we see. More importantly, we can leverage this to our advantage to find all sorts of things that a wordlist wouldn't. Researching as a whole - especially in the context of Cybersecurity encapsulates almost everything you do as a pentester. [MuirlandOracle](#) has created a [fantastic room](#) on learning the attitudes towards how to research, and what information you can gain from it exactly.

"Search Engines" such as Google are huge indexers - specifically, indexers of content spread across the World Wide Web.

These essentials in surfing the internet use "Crawlers" or "Spiders" to search for this content across the World Wide Web, which I will discuss in the next task.

Answer the questions below

Roger dodger!

No answer needed

✓ Correct Answer

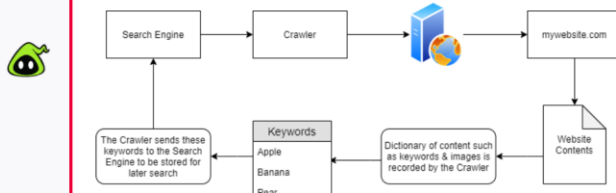
Task 2 Let's Learn About Crawlers

What are Crawlers and how do They Work?

These crawlers discover content through various means. One being by pure discovery, where a URL is visited by the crawler and information regarding the content type of the website is returned to the search engine. In fact, there are lots of information modern crawlers scrape – but we will discuss how this is used later. Another method crawlers use to discover content is by following any and all URLs found from previously crawled websites. Much like a virus in the sense that it will want to traverse/spread to everything it can.

Let's Visualise Some Things...

The diagram below is a high-level abstraction of how these web crawlers work. Once a web crawler discovers a domain such as **mywebsite.com**, it will index the entire contents of the domain, looking for keywords and other miscellaneous information - but I will discuss this miscellaneous information later.



Answer the questions below

Name the key term of what a "Crawler" is used to do. This is known as a collection of resources and their locations

What is the name of the technique that "Search Engines" use to retrieve this information about websites?

What is an example of the type of contents that could be gathered from a website?

Room progress (18%)

Task 3 Enter: Search Engine Optimisation

Search Engine Optimisation

Search Engine Optimisation or SEO is a prevalent and lucrative topic in modern-day search engines. In fact, so much so, that entire businesses capitalise on improving a domains SEO "ranking". At an abstract view, search engines will "prioritise" those domains that are easier to index. There are many factors in how "optimal" a domain is - resulting in something similar to a point-scoring system.

To highlight a few influences on how these points are scored, factors such as:

- How responsive your website is to the different browser types i.e. Google Chrome, Firefox and Internet Explorer - this includes Mobile phones!
- How easy it is to crawl your website (or if crawling is even allowed ...but we'll come to this later) through the use of "Sitemaps"
- What kind of keywords your website has (i.e. In our examples if the user was to search for a query like "Colours" no domain will be returned - as the search engine has not (yet) crawled a domain that has any keywords to do with "Colours"

There is a lot of complexity in how the various search engines individually "point-score" or rank these domains - including vast algorithms. Naturally, the companies running these search engines such as Google don't share exactly how the hierarchic view of domains ultimately ends up. Although, as these are businesses at the end of the day, you can pay to advertise/boost the order of which your domain is displayed.

There are various online tools - sometimes provided by the search engine providers themselves that will show you just how optimised your domain is. For example, let's use [Google's Site](#)

Room progress (18%)

Largest Contentful Paint 12.9 s Cumulative Layout Shift 0.043

Core Web Vitals assessment. To learn more, see [Web Vitals](#).

According to this tool, TryHackMe has an SEO rating of **85/100** (as of 14/11/2020). That's not too bad and it'll show the justifications as to how this score was calculated below on the page.

But...Who or What Regulates these "Crawlers"?

Aside from the search engines who provide these "Crawlers", website/web-server owners themselves ultimately stipulate what content "Crawlers" can scrape. Search engines will want to retrieve **everything** from a website - but there are a few cases where we wouldn't want **all** of the contents of our website to be indexed! Can you think of any...? How about a secret administrator login page? We don't want **everyone** to be able to find that directory - especially through a google search.

Introducing Robots.txt...

Answer the questions below

Use the same SEO checkup tool ([opens in new tab](#)) and other online alternatives to see how their results compare for <https://tryhackme.com> and <http://googledorking.cmnic.co.uk> ([opens in new tab](#))

No answer needed

✓ Correct Answer

Room progress (18%)

Task 4 Beepboop - Robots.txt

Robots.txt

Similar to "Sitemaps" which we will later discuss, this file is the first thing indexed by "Crawlers" when visiting a website.

But what is it?

This file must be served at the root directory - specified by the webserver itself. Looking at this files extension of **.txt**, its fairly safe to assume that it is a text file.

The text file defines the permissions the "Crawler" has to the website. For example, what type of "Crawler" is allowed (I.e. You only want Google's "Crawler" to index your site and not MSN's). Moreover, Robots.txt can specify what files and directories that we do or don't want to be indexed by the "Crawler".

A very basic markup of a Robots.txt is like the following:

```
1 User-agent: *
2 Allow: /
3
4 Sitemap: http://mywebsite.com/sitemap.xml
5
6
```

Here we have a few keywords...

Room progress (18%)

Answer the questions below

Where would "robots.txt" be located on the domain "ablog.com"

Check ?

If a website was to have a sitemap, where would that be located?

Check

How would we only allow "Bingbot" to index the website?

Check

How would we prevent a "Crawler" from indexing the directory "/dont-index-me/"?

Check

What is the extension of a Unix/Linux system configuration file that we might want to hide from "Crawlers"?

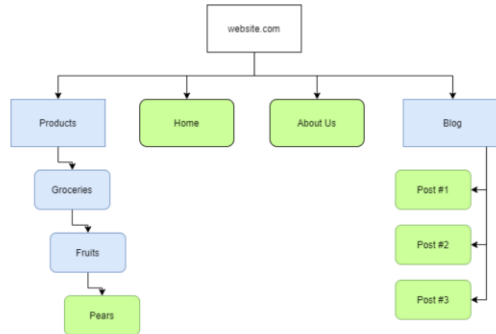
Check ?

Task 5 Sitemaps

Sitemaps

Comparable to geographical maps in real life, "Sitemaps" are just that - but for websites!

"Sitemaps" are indicative resources that are helpful for crawlers, as they specify the necessary routes to find content on the domain. The below illustration is a good example of the structure of a website, and how it may look on a "Sitemap":



The above website structure is more optimized for the search engines.

Answer the questions below

What is the typical file structure of a "Sitemap"?

What real life example can "Sitemaps" be compared to?

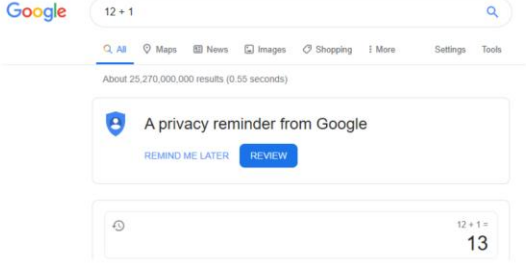
Name the keyword for the path taken for content on a website

Room progress (18%)

Task 6 What is Google Dorking?

Using Google for Advanced Searching

As we have previously discussed, Google has a lot of websites crawled and indexed. Your average Joe uses Google to look up Cat pictures (I'm more of a Dog person myself...). Whilst Google will have many Cat pictures indexed ready to serve to Joe, this is a rather trivial use of the search engine in comparison to what it can be used for. For example, we can add operators such as that from programming languages to either increase or decrease our search results - or perform actions such as arithmetic!



Say if we wanted to narrow down our search query, we can use quotation marks. Google will interpret everything in between these quotation marks as exact and only return the results of the exact phrase provided...Rather useful to filter through the rubbish that we don't need as we have done so below:

Answer the questions below

What would be the format used to query the site `bbc.co.uk` about flood defences

Check ?

What term would you use to search by file type?

Check

What term can we use to look for login pages?

Check ?